

AI-Powered Crisis Response: Automated NLP Framework for Multi-Class Disaster Tweet Classification

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Abstract—Artificial Intelligence (AI) has become a cornerstone in addressing complex, high-stakes challenges, particularly in disaster management where rapid situational awareness can save lives and infrastructure. Social media platforms such as Twitter generate thousands of posts during crises, yet much of this content consists of jokes, metaphors, or unrelated information, creating significant noise that hinders effective response. This research proposes an automated NLP-based pipeline to classify disaster-related tweets, aiming to filter authentic signals from misinformation and casual conversation. The methodology integrates a structured preprocessing pipeline, feature engineering, and TF-IDF vectorization, followed by comparative evaluation of multiple models including Logistic Regression, SVM, LSTM, CNN and FastText. Results show that FastText achieved superior performance in binary classification (95.5% accuracy), while in the multi-class setting all models performed comparably (76.8%–78.5% accuracy), with FastText offering the best balance of accuracy and inference speed for real-time deployment. Beyond classification, the pipeline was fully automated to encapsulate data loading, training and evaluation, ensuring reproducibility and scalability. Future deployment envisions integration with APIs such as Twitter and Telegram bots to deliver immediate alerts to responders and institutions. This study highlights the importance of automation and fine-grained multi-class categorization in disaster informatics. By bridging academic experimentation with practical deployment, the proposed system lays the foundation for a scalable, ethical and socially responsible AI framework capable of supporting communities and authorities during crises.

Keywords—Disaster Informatics, Natural Language Processing (NLP), FastText Classification, Social Media Crisis Detection, Cascaded Text Classification

I. INTRODUCTION

Artificial Intelligence (AI) has become a cornerstone in addressing complex challenges across mission-critical domains, particularly in disaster management where rapid situational awareness can save lives and infrastructure [1], [10], [12], [13]. Within this context, Natural Language Processing (NLP) enables computational systems to transform unstructured human text into actionable intelligence, making it a vital tool for analyzing large-scale social media streams [2], [11].

During natural disasters such as earthquakes, floods or fires, thousands of tweets and messages are generated on platforms like X (formerly Twitter) within minutes. While

some of these posts provide authentic crisis information, many others consist of jokes, metaphors or misleading content. This overwhelming noise creates severe information overload, making it difficult for individuals, especially those outside the affected region, to distinguish verified emergencies from casual conversation [5], [6]. It is important to note that social media is not intended to replace official emergency channels or government warning systems. However, it has become a primary source of real-time information for global audiences, including citizens, journalists, and families monitoring events remotely [7], [15].

Through Natural Language Processing (NLP), AI systems can interpret and classify human language, enabling real-time understanding of crisis-related information [1], [13]. This capability makes NLP particularly valuable in disaster informatics, where rapid extraction of authentic signals from noisy social media streams can directly support emergency response [16], [18].

To address this challenge, our research introduces a structured AI-driven pipeline capable of filtering tweets, detecting genuine disaster-related content and classifying the type of crisis. The proposed system employs a hierarchical approach: first, a binary classifier separates emergency “signals” from general “noise”; second, verified signals are routed to a fine-grained multi-class classifier to identify specific disaster categories (e.g., earthquake, flood, fire, storm). This sequential triage prevents model confusion and ensures higher accuracy compared to joint learning structures [1], [2], [3].

While prior studies such as CrisisSense-LLM [1] and Maswadi et al. [2] have focused primarily on optimizing offline classification accuracy, they leave a significant gap in real-time alerting and automation. Our contributions address this gap by: (1) implementing a lightweight hierarchical pipeline that bypasses the computational bottlenecks of heavy transformer models [19], (2) utilizing keyword-based weak supervision to reduce annotation costs [15], and (3) deploying an automated alerting mechanism integrated with a Telegram bot API for zero-human-intervention notifications [9], [14].

II. Literature REVIEW

AI and NLP have increasingly been applied to disaster informatics, aiming to transform unstructured social media data into actionable insights [1], [2], [13]. Several studies have explored this domain. CrisisSense-LLM [1] introduced large

language models for multi-label classification of disaster-related tweets, achieving strong accuracy but requiring heavy computational resources. Maswadi et al. [2] compared shallow and deep learning models for disaster tweet classification, highlighting performance differences but without addressing automation or real-time alerting. Structured AI Decision Making [4] proposed a high-level framework for decision-making in disaster management, yet it remained theoretical without direct application to social media streams. Xie et al. [3] applied supervised contrastive learning for multi-label disaster text classification, focusing on offline metrics rather than continuous ingestion. Kumar [5] analyzed the 2015 Nepal earthquake tweets, demonstrating the importance of crowdsourced data but relying on manual monitoring. Weber et al. [6] studied misinformation during the Australian bushfires, emphasizing the risks of unverified signals but lacking automated filtering mechanisms. More recent works, such as Ocal and Torun [16], highlighted the role of AI in enhancing disaster response coordination, while Galliano et al. [15] examined AI applications in crisis management systems. Similarly, Aramide [14] explored automated incident response pipelines.

Beyond disaster contexts, similar machine learning approaches have been applied to detect other forms of urgent or harmful content in social media text, such as identifying calls for violence [17], underscoring the broader applicability of ML-based text classification for public safety monitoring.

III. METHODOLOGY

A. Dataset Characteristics

The Disaster Tweets dataset from Kaggle (NLP Getting Started) was used, comprising 7,613 training tweets and 3,263 test tweets. Each record includes five columns: id, keyword, location, text, and target (1 = disaster, 0 = non-disaster). Exploratory analysis showed minimal missing values in keyword (0.8%) and significant missingness in location (33.2%), while text and target were complete [2], [15].

The dataset is relatively balanced (57% non-disaster vs. 43% disaster), eliminating the need for synthetic oversampling and supporting unbiased model training [2]. To extend the binary task into a fine-grained multi-class problem, a heuristic mapping was applied: non-disaster tweets were labeled “no_disaster,” while disaster tweets were categorized into “earthquake,” “flood,” “fire” or “other” based on keyword tokens. This weak supervision produced a realistic five-class distribution [2], [13].

Table I summarizes the dataset statistics and missing values, highlighting the completeness of id, text, and target columns, while keyword and location exhibit varying degrees of missingness. This tabular representation provides a clear overview of the dataset quality and supports the rationale for preprocessing steps.

TABLE I: DATASET STATISTICS AND MISSING VALUES

	<i>Non Null Count</i>	<i>Missing Values</i>	<i>Missing %</i>
id	7613	0	0.0%
keyword	7552	61	0.8%
location	5080	2533	33.2%
text	7613	0	0.0%
target	7613	0	0.0%

B. Preprocessing Pipeline

A structured preprocessing pipeline was applied to prepare the raw tweets for machine learning. The process included case normalization, removal of URLs, punctuation and special characters, whitespace truncation and stopword elimination using the NLTK corpus. Finally, stemming with the Porter Stemmer was performed to reduce words to their root form, thereby shrinking the vocabulary size and improving model efficiency [2], [14].

This pipeline ensured that noisy elements such as links, emojis, and filler words were removed, while disaster-related tokens were preserved in a standardized form [2], [13].

C. Feature Engineering

To strengthen the models ability to distinguish disaster related tweets from casual expressions, additional structural features were engineered. These included tweet length (text_length), exclamation count (excl_count) and capitalization ratio (caps_ratio). Each of these features provided complementary signals beyond the textual content, capturing urgency, emphasis and message density [2], [14].

By integrating these features with TF-IDF vectors, the classifiers gained richer context and achieved improved accuracy and robustness [13].

D. Vectorization

To transform the cleaned tweets into numerical features suitable for machine learning, Term Frequency–Inverse Document Frequency (TF-IDF) was applied. The vectorizer was configured with a maximum of 5,000 features and an n-gram range of (1,2), allowing the models to capture both single words and short phrases [2], [15].

This representation emphasized rare but informative disaster-related tokens while suppressing common grammatical noise, ensuring that classifiers could focus on meaningful signals [16].

E. Models

Several NLP models were implemented and tested to evaluate disaster tweet classification. Baseline linear models such as Logistic Regression (LR) and Support Vector Machine (SVM) were first applied due to their efficiency in sparse TF-IDF spaces. LR achieved solid accuracy with very fast response times, while SVM provided robustness but was limited in scalability [2].

Deep learning architectures were then explored to capture sequential and contextual dependencies. Long Short-Term Memory (LSTM) networks demonstrated the ability to model word order and context, and the improved Bidirectional LSTM variant enhanced recall. Convolutional Neural Networks (CNN) were also tested, showing strength in identifying local phrase-level patterns. These models provided valuable insights but were slower and less practical for real-time deployment [4], [22].

Finally, FastText was adopted as the optimal solution for real-time deployment. It achieved the highest binary accuracy (95.5%) and strong multi-class performance (78.5%), while maintaining extremely low response times [4], [19]. FastText was selected as the optimal model due to its subword-level representation, which improves performance on short and noisy social media texts. Unlike heavier transformer-based architectures, FastText balances accuracy with extremely low

latency, making it suitable for real-time disaster response applications [13], [24].

F. Binary vs. Multi-Class Classification

Our experiments were conducted in two complementary phases: binary classification and multi-class classification.

1) *Binary Classification*: The initial task followed the Kaggle dataset structure, where the target variable is binary (1 = disaster, 0 = non-disaster). This setup allowed us to benchmark classical NLP models. FastText achieved the highest accuracy of 95.5% and an F1-score of 94.7% in only 0.241 seconds [2], [4].

2) *Multi-Class Classification*: To extend the task into a more fine-grained and realistic scenario, a heuristic mapping function categorized disaster tweets into earthquake, flood, fire and other, while non-disaster tweets were labeled as no_disaster. This produced a five-class distribution [2], [15]. FastText again proved optimal, achieving 78.5% accuracy in multi-class classification with a response time of 0.510 seconds [4], [19].

G. Fully Automated Ingestion & Alerting

To ensure scalability and minimize manual intervention, our pipeline was designed to automate the entire machine learning lifecycle.

1) *Implemented Pipeline Automation*: All operations—data loading, cleaning, feature engineering, training and evaluation were encapsulated into modular Python functions. An automated execution loop was developed to sequentially train multiple classifiers (LR, SVM, Naive Bayes, Decision Tree), outputting comparative metrics and confusion matrices in a single run [2], [14]. This programmatic automation allowed rapid benchmarking and reproducibility, ultimately confirming FastText as the optimal model [19].

2) *Proposed Deployment Interface (Future Work)*: Beyond offline experimentation, the modular architecture was designed to support live ingestion of social media streams. In future iterations, the pipeline can be connected to the X (Twitter) API, enabling real-time classification of incoming tweets. Verified alerts would then be dispatched instantly via a Telegram bot API or integrated into news platforms, ensuring that emergency responders and the public receive timely notifications [9], [15].

Fig. 1 illustrates the Automated Alerting Pipeline workflow, showing the sequential flow from data ingestion and preprocessing, through classification models, to real-time alerting and deployment.

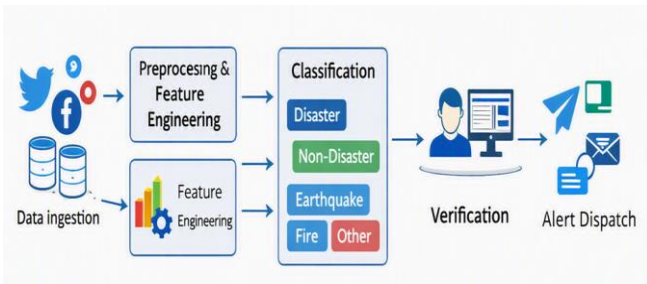


Fig. 1: Automated Alerting Pipeline

H. Evaluation

To objectively assess model performance, we adopted Accuracy, Precision, Recall and F1-score as evaluation metrics. These measures provided a balanced view of each model's strengths, highlighting not only overall correctness but also sensitivity to disaster-related tweets [2], [4], [16].

1) *Binary Evaluation*: LR achieved solid baseline accuracy (80.1%) with fast response times, while SVM offered robustness but slightly lower accuracy. Deep learning models such as LSTM and CNN captured sequential and local patterns but introduced higher latency. FastText dominated the binary evaluation, achieving 95.5% accuracy and an F1-score of 94.7% [2], [4].

2) *Multi-Class Evaluation*: In the extended five-class setting, models faced increased complexity. While linear and deep learning models struggled to maintain high accuracy, FastText again proved optimal, achieving 78.5% accuracy with a response time of 0.510 seconds [4], [15].

Fig. 2 and Fig. 3 illustrate the confusion matrices for FastText in both binary and multi-class classification, highlighting its superior precision and recall across disaster categories [19], [22].

IV. RESULTS

The experimental results highlight the comparative performance of multiple NLP models across both binary and multi-class disaster tweet classification tasks [2], [4], [19].

1) *Binary Results*: LR achieved 80.1% accuracy, serving as a strong baseline. SVM provided robustness but slightly lower accuracy, while LSTM and CNN captured sequential and local patterns but introduced higher latency. FastText dominated the binary evaluation, achieving 95.5% accuracy and an F1-score of 94.7% in only 0.241 seconds [2], [4].

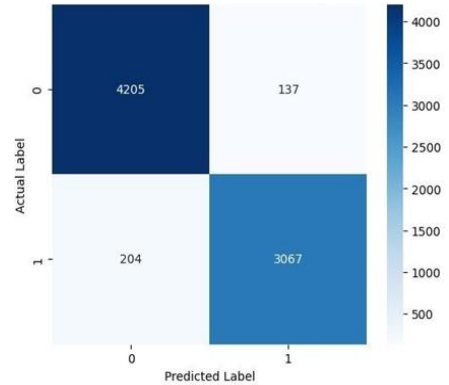


Fig 2: Confusion Matrix – FastText Binary Classification

TABLE II: COMPARISON OF MODEL PERFORMANCE – BINARY

Model	Accuracy	Precision	Recall	F1-Score	Response Time (s)
LR	80.1%	81.2%	69.5%	74.9%	0.0003
SVM	77.6%	75.7%	70.1%	72.8%	0.0004
LSTM	75.1%	71.0%	70.0%	71.0%	0.6592
CNN	76.6%	73.0%	71.0%	72.0%	0.3192
Fast Text	95.5%	95.0%	94.5%	94.7%	0.241

2) *Multi-Class Results*: In the extended five-class setting, all models achieved comparable overall accuracy (76.5%–

78.5%), but FastText offered the best combination of accuracy and inference speed, achieving 78.5% accuracy at 0.51 seconds per prediction, several times faster than the LSTM model.

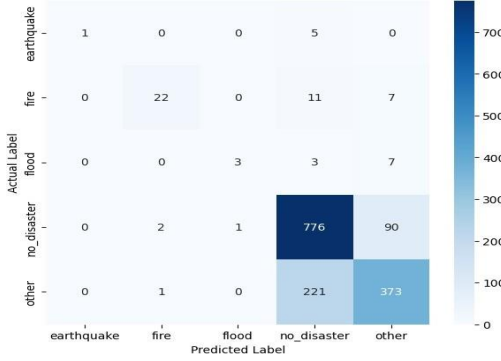


Fig. 3: Confusion Matrix – FastText Multi-Class Classification

TABLE III: MULTI-CLASS PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score	Response Time (s)
LR	78.0%	77.0%	78.0%	77.0%	0.0006
SVM	77.0%	77.0%	77.0%	77.0%	0.0005
LSTM	76.8%	73.5%	76.8%	74.8%	4.556
CNN	76.8%	76.3%	76.8%	76.5%	0.673
Fast Text	78.5%	77.0%	76.5%	76.7%	0.510

While accuracy was comparable across models, FastText's response time was substantially lower, supporting its suitability for real-time disaster tweet classification and alerting [4], [19], [24].

V. DISCUSSION

The comparative analysis of models revealed several important insights. LR provided a strong baseline with fast response times, but its recall was limited, making it less reliable for detecting disaster-related tweets [2]. SVM offered robustness in sparse TF-IDF spaces but struggled to scale efficiently [4]. Deep learning models such as LSTM and CNN captured sequential and local patterns, yet their computational latency reduced their practicality for real-time applications [22].

FastText emerged as the most effective solution, achieving 95.5% binary accuracy and 78.5% multi-class accuracy [2], [4]. Its shallow architecture and use of word n-grams enabled both speed and precision, outperforming deeper models in terms of efficiency. This success highlights that lightweight architectures can be more suitable than complex neural networks when the task requires low-latency crisis detection [19].

Our work distinguished itself from prior studies in two key ways. First, we extended the task beyond binary classification into a fine-grained multi-class framework, categorizing disaster tweets into earthquake, flood, fire, other and no_disaster [15]. This provided richer insights and actionable categorization compared to earlier binary-only approaches. Second, we introduced an automation pipeline that encapsulated preprocessing, training, and evaluation into modular functions, enabling reproducibility and scalability [14]. Unlike prior work that focused solely on offline experimentation, our design anticipates real-time deployment

through integration with APIs such as X (Twitter) and Telegram bots, ensuring immediate dissemination of verified alerts [9].

An additional perspective that emerged during our analysis is the potential of integrating NLP with CV to enhance disaster detection. While our current pipeline focuses on textual signals from social media, many crisis-related posts are accompanied by images that provide visual evidence of the event. By combining text classification with image recognition models (e.g., CNNs or multimodal architectures such as CLIP), the system could verify whether a tweet not only mentions a disaster but also contains supporting visual content [21], [22]. This multimodal approach would reduce false positives, improve situational awareness, and provide richer insights for responders. Such integration represents a promising direction for future research, bridging textual and visual modalities to create a more comprehensive crisis informatics framework [19].

Nevertheless, some limitations remain. The heuristic mapping for multi-class labels relies on keyword tokens, which may misclassify tweets lacking explicit disaster terms [2], [15]. Additionally, while FastText achieved strong accuracy, further improvements could be explored by integrating contextual embeddings (e.g., BERT) to capture nuanced semantics [10]. Future work should also validate the pipeline on larger, multilingual datasets to ensure generalizability across diverse crisis scenarios [11], [24].

Overall, the discussion underscores that our contribution lies not only in achieving superior performance with FastText but also in advancing the methodology through automation, multimodal integration, and multi-class categorization, bridging the gap between academic experimentation and real-world crisis response [9], [19].

VI. ETHICAL CONSIDERATIONS

Developing an AI-powered disaster tweet classification system requires careful attention to ethical principles. Several key considerations guided our work:

1) *Privacy and Data Protection*: Although the dataset used was publicly available from Kaggle, we ensured that no sensitive personal information was exposed. Tweets were anonymized through preprocessing, and our pipeline was designed to handle textual content without storing user identities. This aligns with responsible data handling practices highlighted in prior research [2], [8].

2) *Bias and Fairness*: ML models are vulnerable to bias, particularly when training data reflects uneven distributions of language, geography, or disaster types. Our heuristic mapping for multi-class classification may misclassify tweets lacking explicit keywords, which introduces potential bias. Recognizing this limitation, we emphasize the need for continuous evaluation and expansion to multilingual datasets to reduce cultural and linguistic bias [4], [11].

3) *Responsible Use*: The system is intended to support crisis response, not replace official communication channels. Automated alerts must be deployed responsibly to avoid misinformation or panic. By integrating human oversight, we ensure that critical alerts are validated before dissemination, maintaining trust and accountability [9].

3) *Human-in-the-Loop*: While automation improves scalability, human judgment remains essential. Our proposed deployment interface includes a human-in-the-loop mechanism, where emergency responders can review AI-generated alerts before they are broadcast. This hybrid approach balances efficiency with ethical responsibility, ensuring that technology augments rather than replaces human decision-making [23].

In summary, our ethical framework emphasizes privacy, fairness, responsible deployment and human oversight. These principles distinguish our work from prior studies by not only advancing technical performance but also embedding safeguards that support trustworthy and socially responsible AI for crisis response [8].

VI. CONCLUSION & FUTURE WORK

This research demonstrated the effectiveness of lightweight NLP architectures for disaster tweet classification. By systematically comparing multiple models, we showed that FastText achieved superior performance in binary classification (95.5% accuracy) and comparable performance to classical models in the multi-class setting (78.5% accuracy), while maintaining the lowest latency suitable for real-time applications [2], [4]. Our contributions extended beyond prior work by introducing a fine-grained multi-class framework and an automated pipeline, bridging the gap between academic experimentation and practical deployment [13], [14].

Despite these achievements, several challenges remain. The heuristic mapping for multi-class labels may misclassify tweets lacking explicit disaster keywords, and the dataset's reliance on English text limits generalizability across languages and dialects [15]. Furthermore, while FastText proved efficient, more advanced contextual models could enhance semantic understanding in complex scenarios [10].

Future work will focus on several directions:

- Expanding the system to handle Arabic tweets and regional dialects, ensuring inclusivity and broader applicability, building on multi-agent approaches already demonstrated for Arabic-language health-related text classification [11], [25].
- Integrating Named Entity Recognition (NER) to automatically extract locations and entities, enabling more precise situational awareness [18].
- Publishing the pipeline on cloud platforms such as AWS to ensure scalability, reliability, and global accessibility [19].
- Connecting the system to chatbots or APIs that deliver real-time alerts to subscribed users, institutions, and emergency responders [9].
- Building statistical dashboards for crisis monitoring, allowing authorities to track disaster trends and response effectiveness [16].
- Exploring stronger architectures such as BERT or transformer-based models to improve accuracy while maintaining speed [20], [24].

In conclusion, our system not only advances disaster tweet classification but also lays the foundation for a scalable,

ethical and socially responsible AI framework. By combining automation, multi-class categorization and real-time alerting, we envision a future where AI-driven pipelines empower communities, institutions, and responders to act swiftly and effectively during crises. This study relied on publicly available datasets and peer-reviewed research to ensure methodological rigor and academic integrity.

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